Semester 1, 2012 Final Exam

Questions on Normalization and SQL with Solutions

Question 22 – Normalization (18 marks)

Use the following description and the three tables to answer parts (A) and (B) of this question.

A transport company provides a number of trucks for cartage purposes. These trucks are hired by people for a number of different reasons, for example, for carting household items when moving from one house to another.

The trucks operate from a base, which is managed by at least one employee of the company. A truck may operate from one or multiple bases over a period of time. A base will have at least one truck at any given time.

The company hired a database designer to create a MS Access database that could be used by the company to store details about the trucks. The implemented MS Access database contains three tables, which are provided together with the relational schema in Figure 1.

Base					
Base	**City**	**State**	**Area Code**	**Phone**	**Manager**
501	Murfreesboro	TN	615	123-4567 or 123-4577	Andrea D. Gallager
502	Lexington	KY	568	234-5678 or 234-5699	Andrea D. Gallager and Maria J. Talindo
503	Cape Girardeau	MO	456	345-6789	Maria J. Talindo
504	Dalton	GA	901	456-7890	Peter F. McAvee

Truck						
Truck #	**Base**	**Base City**	**Base Phone**	**Truck Type**	**Miles**	**Serial Num**
1001	501	Murfreesboro	123-4567 or 123-4577	1	32123.5	AA-322-12212-W11
1002	501	Murfreesboro	123-4567 or 123-4577	1	76984.3	AC-342-22134-Q23
1002	502	Lexington	234-5678 or 234-5699	1	76984.3	AC-342-22134-Q23
1003	501	Murfreesboro	123-4567 or 123-4577	2	12346.6	AC-445-78656-Z99
1003	503	Cape Girardeau	345-6789	2	12346.6	AC-445-78656-Z99
1005	503	Cape Girardeau	345-6789	2	45673.1	FR-998-32245-W12
1006	501	Murfreesboro	123-4567 or 123-4577	2	93245.7	AD-456-00845-R45
1006	503	Cape Girardeau	345-6789	2	93245.7	AD-456-00845-R45

Truck Type	
Type	**Description**
1	Single box, double-axle
2	Single box, single-axle
3	Tandem trailer, single-axle

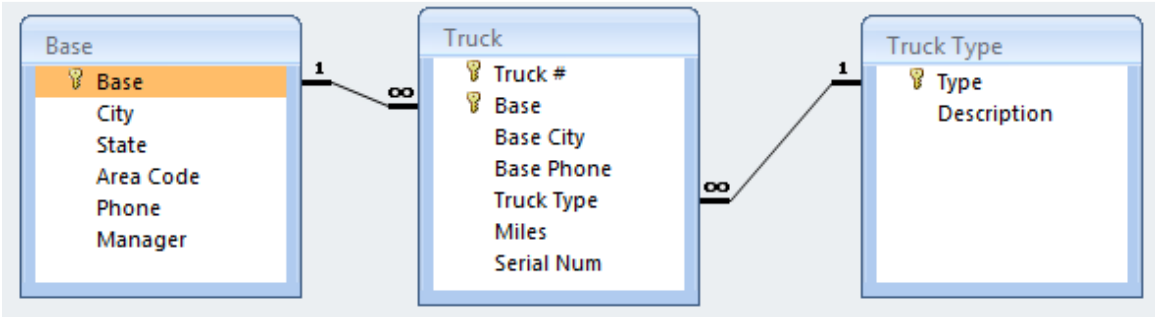


Figure 1

(A) Discuss some problems with the design of above database and provide examples of the problems that you have identified. (4 marks).

Ans:

1. Redundancy:
    a. The truck details are repeating the truck table because a truck can operate from multiple bases.
    b. The base details are repeating in base and truck tables.
2. Multi-valued attributes: Base phone and Base Manager are both mult-values attributes
3. Attribute Naming: The names of the attributes do not follow a particular format.

(B) The current database has three tables. Redesign, that is normalize, the database so that all the tables are in 3NF and create the dependency diagrams for all the tables in your re-designed database. Use the primary keys identified in the relational schema for the normalization process. (14 marks)

Employee (**Emp_Num**, Emp_FName, Emp_LName)

Base (**Base_Num**, Base_City, Base_State, Base_Areacode)

Base_Phone (**Base_Num**, **Base_Phone_Num**)

Base_Manager (**Base_Num**, **Emp_Num**)

Truck_Type (**Type_ID**, Type_Desc)

Truck (**Truck_Num**, Truck_Serial, Truck_Miles, Type_ID)

Base_Assign (**Truck_Num**, **Base_Num**)

Question 24 – Structured Query Language (17 marks)

Use the following description and the ERD given below (see Figure 2) to answer parts (A) to (E) of this question. Write SQL statements to answer each part and your SQL statement should be written for a MySQL database.

You are given a database that is used by a video library to store data about movies, copies of each movie, borrowers, and renting of the movies. A movie can have one or more copies. Each copy can be rented by one of more borrowers.

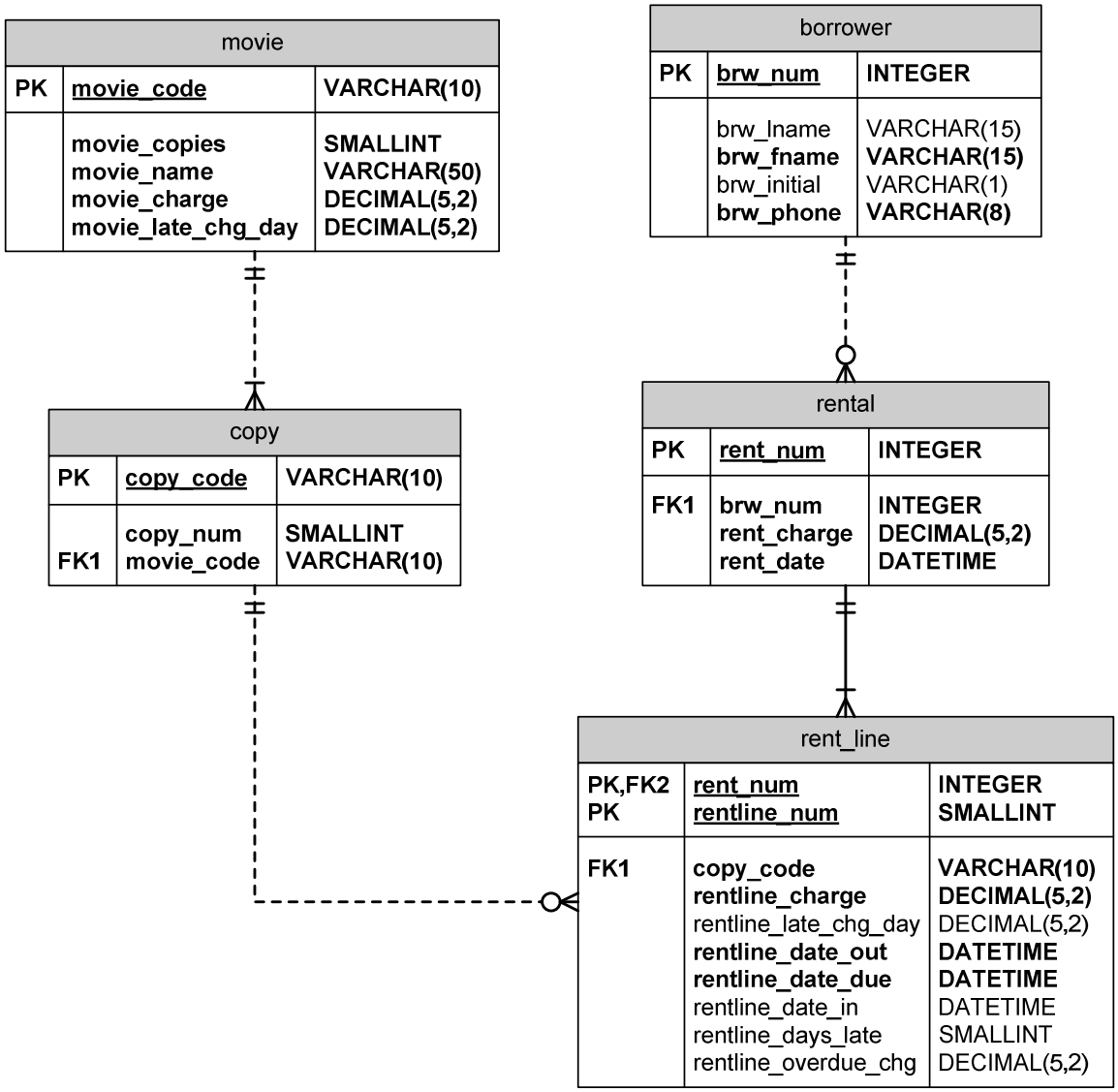


Figure 2

(A) List the details of all borrowers, sorted by their last name in increasing order. (2 marks)

```
select *

from borrower

order by brw_lname;
```

(B) List the number and names of all borrowers that have not borrowed any movies. (3 marks)

```
select b.brw_num,

   concat_ws(" ", brw_fname, brw_lname) as 'Name'

from borrower as b

where b.brw_num not in(select distinct(brw_num) from

rental);
```

(C) List the code(s), name(s) and the price of the most expensive movie(s) [Hint: use the movie_charge column]. (3 marks)

```
select movie_code, movie_name, movie_charge

from movie

where movie_charge in

   (select max(movie_charge) from movie);
```

(D) List the movie code, name and the number of copies of each movie that has more than two copies. (4 marks)

```
select m.movie_code, movie_name, count(copy_code) as 'copies'
from movie as m, copy as c
where m.movie_code = c.movie_code
group by m.movie_code, movie_name
having count(copy_code) > 2;
```

```
select m.movie_code, movie_name, count(copy_code) as 'copies'
from movie as m, copy as c
where m.movie_code = c.movie_code
group by m.movie_code, movie_name
having copies > 2;
```

```
select movie_code, movie_name, count(copy_code) as 'copies'
from movie natural join copy
group by movie_code, movie_name
having count(copy_code) > 2;
```

(E) List the movie code and name of all movies where at least one copy of the movie has never been borrowed. (5 marks)

```
select movie_code, movie_name
from movie
where movie_code in
(select distinct movie_code from copy where copy_code not in
(select distinct copy_code from rent_line));
```

```
SELECT  distinct  movie_code,movie_name

FROM movie NATURAL JOIN copy

WHERE copy_code NOT IN

      (SELECT distinct(copy_code) FROM rent_line);
```

Semester 1, 2011Final Exam

Questions on Normalization and SQL with Solutions

Question 22 – Normalization (12 marks)

Use the following description and the two tables to answer parts (A) to (C) of this question.

A small software development firm uses the following two tables to store details about the software development projects the firm undertakes, the employees who worked on the projects, and the project manager. For each employee that worked on the project, the employee's details, the dates when the employee worked on the project, and the total hours, hourly rate and total payment is recorded for each day the employee worked on a particular project. For the project manager, only the manager's employee number is recorded as the Manager #, and the manager's name is not recorded.

The employee details are stored in a separate table. The two tables are shown below. The first table is the Project table, while the second one is the Employee table.

Project								
Project Number	**Title**	**Emp #**	**Name**	**Work Date**	**Hours Worked**	**Hourly Rate**	**Total Payment**	**Manager #**
15	Evergreen	103	June Arbough	04-Mar-04	2.60	$84.50	$219.70	105
15	Evergreen	101	John News	05-Mar-04	3.60	$105.00	$378.00	105
15	Evergreen	103	June Arbough	05-Mar-04	1.90	$84.50	$160.55	105
15	Evergreen	105	Alice Johnson	05-Mar-04	2.00	$105.00	$210.00	105
15	Evergreen	102	David Senior	06-Mar-04	3.80	$96.75	$367.65	105
15	Evergreen	101	John News	06-Mar-04	2.30	$105.00	$241.50	105
15	Evergreen	103	June Arbough	06-Mar-04	3.00	$84.50	$253.50	105
22	Rolling Tide	113	Delbert Joenbrood	05-Mar-04	2.50	$48.10	$120.25	113
22	Rolling Tide	105	Alice Johnson	05-Mar-04	5.20	$105.00	$546.00	113
22	Rolling Tide	104	Anne Ramoras	06-Mar-04	2.60	$96.75	$251.55	113
22	Rolling Tide	111	Geoff Wabash	06-Mar-04	4.00	$26.87	$107.48	113
22	Rolling Tide	105	Alice Johnson	07-Mar-04	2.70	$105.00	$283.50	113
22	Rolling Tide	106	William Smithfield	07-Mar-04	2.40	$35.75	$85.80	113
25	Starflight	101	John News	05-Mar-04	1.70	$105.00	$178.50	101
25	Starflight	114	Annelise Jones	06-Mar-04	1.80	$48.10	$86.58	101
25	Starflight	114	Annelise Jones	07-Mar-04	3.40	$48.10	$163.54	101

Employee					
Emp #	**Name**	**Hire Date**	**Job Code**	**Job Desc**	**Job Hourly Rate**
106	William Smithfield	22-Jun-03	500	Programmer	$35.75
107	Maria Alonzo	10-Oct-91	500	Programmer	$35.75
102	David Senior	12-Jul-87	501	Systems Analyst	$96.75
104	Anne Ramoras	15-Nov-85	501	Systems Analyst	$96.75
108	Ralph Washington	22-Aug-89	501	Systems Analyst	$96.75
109	Larry Smith	18-Jul-95	501	Systems Analyst	$96.75
115	Travis Bawangi	25-Jan-90	501	Systems Analyst	$96.75
101	John News	08-Nov-98	502	Database Designer	$105.00
105	Alice Johnson	01-Feb-91	502	Database Designer	$105.00
103	June Arbough	01-Dec-94	503	Electrical Engineer	$84.50
110	Gerald Olenko	11-Dec-93	505	Civil Engineer	$55.78
111	Geoff Wabash	04-Apr-89	506	Clerical Support	$26.87
112	Darlene Smithson	23-Oct-92	507	DSS Analyst	$45.95
113	Delbert Joenbrood	15-Nov-94	508	Applications Designer	$48.10
114	Annelise Jones	20-Aug-91	508	Applications Designer	$48.10
117	Angie Williamson	19-Jun-94	509	Bio Technician	$34.55
116	Gerald Pratt	05-Mar-95	510	General Support	$18.36
118	James Frommer	04-Jan-04	510	General Support	$18.36

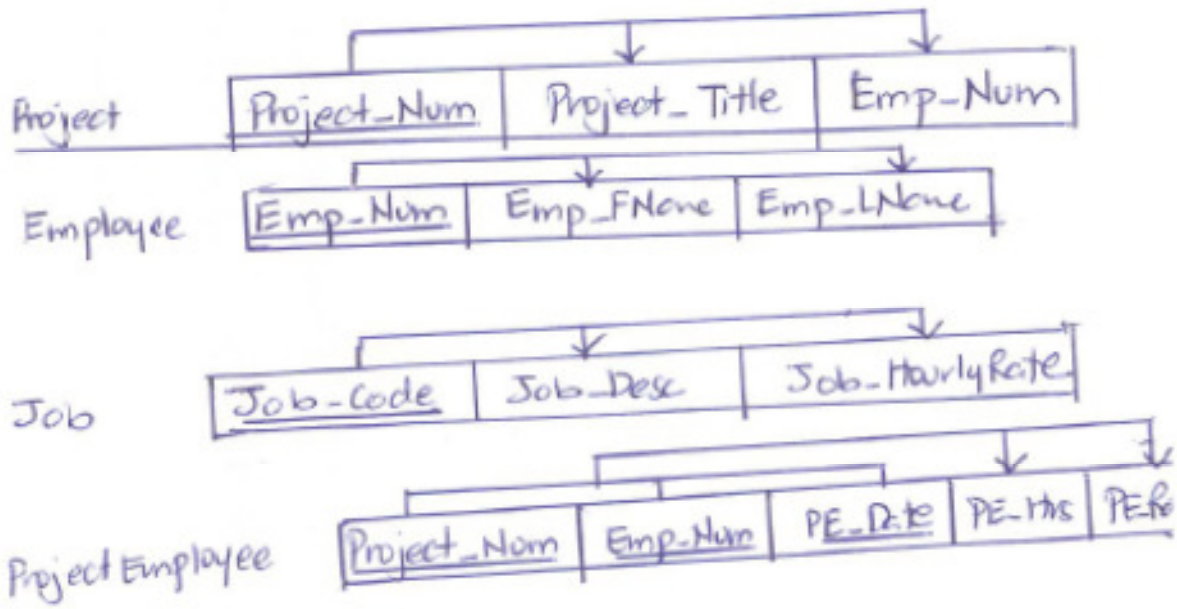
(A) When is normalization process used? (2 marks)

When you are required to analyze and correct an existing database.

(B) The current database has two tables. Redesign, that is normalize, the database so that all the tables are in 3NF and create the dependency diagrams for all the tables in your re-designed database. A database designer has identified the following primary keys for the above two tables:

Table	**Primary Key**
Project	Project Num, Emp #, Work Date
Employee	Emp #

Use these primary keys for the normalization process. (8 marks)



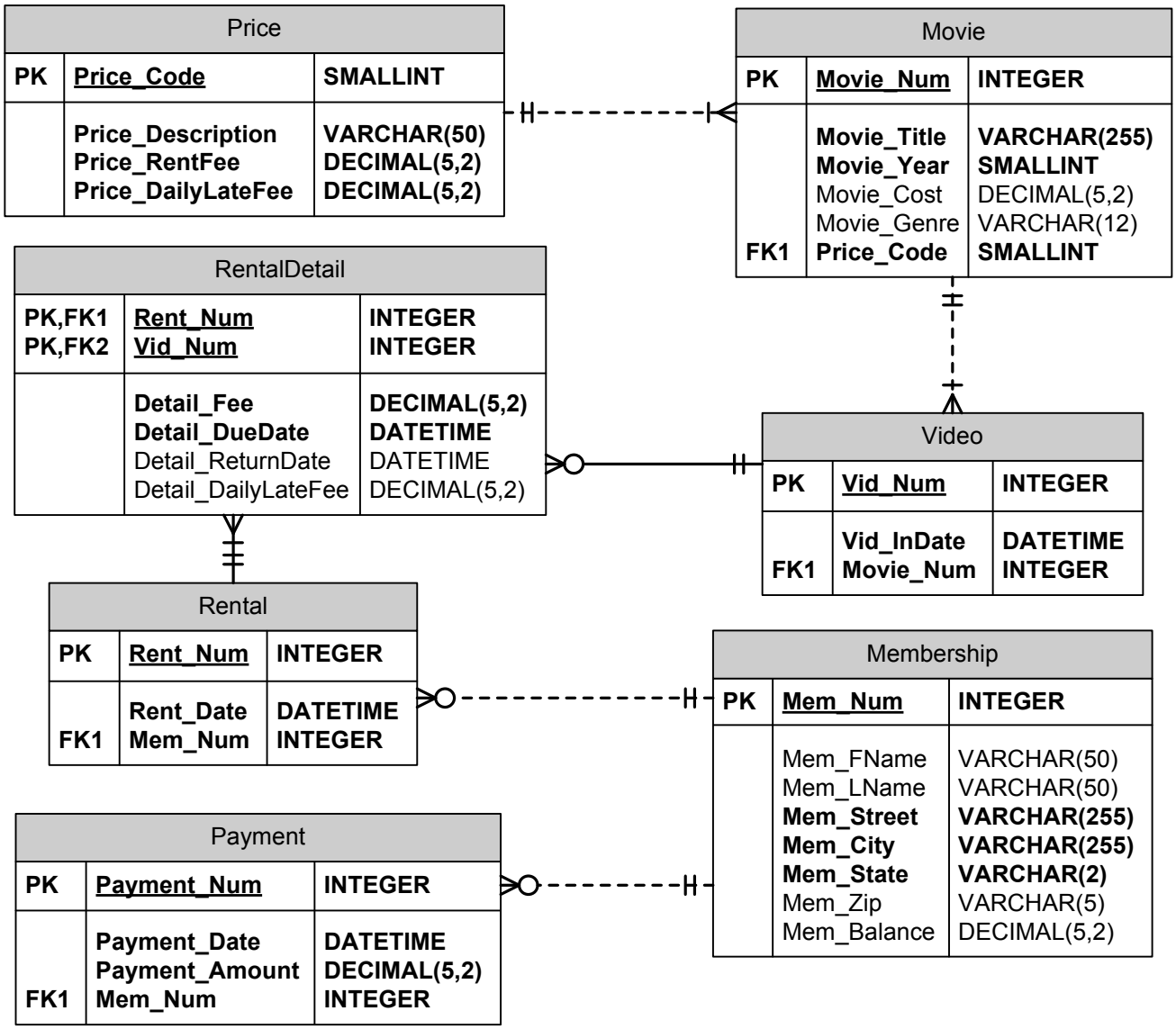
(C) Briefly explain when 4NF is violated. (2 marks)

Fourth normal form is violated when the table has multiple independent multi-valued attributes. However, these multi-valued attributes can be determined by a common key.

Question 23 – Structured Query Language (28 marks)

Use the following description and the ERD given below to answer parts (A) to (I) of this question. Write SQL statements to answer each part and your SQL statement should be written for a MySQL database.

This database is used by a video library to store data about movies, copies of each movie (video), membership, and rental records of the movies. A movie can have one or more copies. Each copy can be rented by one or more members. The movie copy record is stored in the Video table. Each movie has a particular price scheme and all price schemes are included in the Price table.



(A) List the details of all members, sorted by their last name in descending order. (2 marks)

```
SELECT *

FROM membership

ORDER BY mem_lname DESC
```

(B) List the number of rental transactions where the video was returned late that is after the due date. (2 marks)

```
SELECT COUNT(*) FROM rentaldetail

WHERE detail_returndate > detail_duedate
```

(C) List the movie number, title, rental fee and late return fee for each movie. Sort the list by rental fee in increasing order. (3 marks)

```
SELECT movie_num, movie_title, price_rentfee, price_dailylatefee

FROM movie NATURAL JOIN price

ORDER BY price_rentfee
```

(D) List the movie number and title of the most expensive movie(s). (3 marks)

```
SELECT movie_num, movie_title

FROM movie NATURAL JOIN price

WHERE price_rentfee IN (SELECT DISTINCT MAX(price_rentfee) from
price)
```

(E) List the number, name (combine first and last names together) and state of all members that live in the same state as members with member numbers 107 and 106. Do not include members 106 and 107 in the output. (5 marks)

```
SELECT mem_num, mem_state, CONCAT( mem_lname, " ", mem_fname ) AS
'member name'
FROM membership
WHERE mem_state IN (
SELECT mem_state
FROM membership
WHERE mem_num IN (107, 106))
AND mem_num NOT IN (107, 106)
```

(F) List the video number, movie number and movie title of all movies that have not been rented out. (3 marks)

```
SELECT vid_num, movie_num, movie_title
FROM movie natural join video
WHERE vid_num NOT IN ( SELECT DISTINCT (vid_num) FROM
rentaldetail )
```

(G) Create a view named *moviesales* that lists the movie number, title and the total amount of sales for each movie. Include those movies that have not been rented. Exclude any late return penalty that may have been incurred as a result of the member returning the movie late. (5 marks)

```
CREATE VIEW moviesales AS
SELECT movie_num, movie_title, sum( detail_fee ) AS 'Sales
Amount'
FROM rentaldetail
RIGHT JOIN (
      movie NATURAL JOIN video
)
USING (vid_num)
GROUP BY movie_num, movie_title
```

(H) Write the UPDATE statement to update the first name of the member with number 106 from Jane to Jene. (2 marks)

```
UPDATE membership SET mem_fname = 'Jene' WHERE mem_num = 106;
```

(I) Create a trigger that will automatically update the members balance when the payment table is updated with the payment records. You can use the following partially complete trigger definition. (3 marks)

```
CREATE TRIGGER memberbalanceupdate

AFTER INSERT ON payment

FOR EACH ROW BEGIN

UPDATE membership SET mem_balance = mem_balance -
NEW.payment_amount

WHERE membership.mem_num = NEW.mem_num;

END
```